Phase 8.1 — BeatAhead ISI Sensitivity & Robustness Audit Report

Document ID: BEATAHEAD-PHASE8-1-SENSITIVITY-AUDIT
Execution Timestamp: 2026-09-22 19:55:00 UTC
Environment: Mathematical Sensitivity & Robustness Audit
Status: COMPLETE (Zero Retraining / Zero Code Modification / Zero Threshold Alteration)
Frozen Parameters Evaluated: - Model Decision Threshold: $\tau = 0.156742$ (Preserved strictly) - Model Evidence Curvature: $\gamma_1 = 1.4$, $\gamma_2 = 4.0$ - Component Weights: $w_{\text{model}} = 0.45$, $w_{\text{auto}} = 0.30$, $w_{\text{perf}} = 0.25$ - Personal Baseline: $\text{ISI}_{\text{base}} = 40.0$, Scale $S = 60.0$, EMA $\alpha = 0.02$ - Target Invariant: $0 \le \text{ISI} \le 100$

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1. Executive Summary

Phase 8.1 conducts an isolated mathematical sensitivity and robustness audit of the prototype **Ischemic Stress Index (ISI)** formula defined in Phase 8 (`docs/ISI_SPECIFICATION.md`).

In strict observance of project boundaries: - The frozen XGBoost model (`models/beatahead_phase5_model.joblib`) was **not** retrained or altered. - The 26-feature Matrix A schema (`models/feature_schema.json`) was **not** modified. - The 100-case VitalDB dataset and the 20-case locked test split were **not** accessed or optimized against. - No website code in `C:\ischemic` was touched. - The calibrated decision threshold ($\tau = 0.156742$) was **preserved exactly**.

Key Audit Findings:

1. **Mathematical Invariant Holds Universally ($0 \le \text{ISI} \le 100$)**: Across 10,000 dense sampling points, all 14 boundary edge cases, and multi-component saturation tests, the composite ISI score remained strictly bounded within $[0, 100]$. Maximum possible theoretical saturation produced $\text{ISI}_{\text{raw}} = 103.35$ (cleanly clamped to 100), and minimum possible theoretical input produced $\text{ISI}_{\text{raw}} = 25.6$ (clamped to 26).
2. **Monotonicity & Exact Threshold Continuity**: The model evidence transformation $E_{\text{model}}(p)$ is strictly monotonically increasing ($\frac{\partial E}{\partial p} > 0$) across the entire domain $p \in [0, 1]$. At the exact decision threshold $p = 0.156742$, $E_{\text{model}} = 0.500000$, ensuring seamless continuity between sub-threshold suppression and post-threshold scaling.
3. **Weight Perturbation Stability**: Perturbing component weights by $\pm 22\%$ (Model: $0.35–0.55$, Autonomic: $0.20–0.40$, Perfusion: $0.15–0.35$) altered normal resting scores by < 3 ISI points and high-stress scores by < 6 ISI points, demonstrating structural robustness without erratic bifurcation.
4. **Baseline Anti-Drift Rule Verified**: Under an acute simulated stress event (HR rising from $68 \rightarrow 88\text{ bpm}$ over 30 minutes), standard un-gated EMA ($\alpha=0.02$) drifted by $+8.5\text{ bpm}$, erroneously normalizing the stress state. The Phase 8 **Gated Anti-Drift Rule** successfully froze baseline updates during elevated states, preserving a true $+29.4\%$ quantified physiological deviation throughout the event.
5. **Parameter Identifiability Reality**: The VitalDB intraoperative development data **cannot** legitimately estimate or validate product-layer constants (w_i, S, γ_i). Anesthetized surgical hemodynamic patterns do not translate to ambulatory wellness indices. All product-layer parameters are formally designated: **"Prototype engineering parameters — not clinically validated."**

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2. Baseline ISI Mathematical Formulation

The baseline prototype specification evaluated in this audit consists of the following closed-form system:

2.1 Model Evidence Mapping

$$E_{\text{model}}(p) = \begin{cases} 0.50 \cdot \left(\frac{p}{\tau} \right)^{\gamma_1} & p < \tau \ 0.50 + 0.50 \cdot \left(1 - \exp\left(-\gamma_2 \frac{p - \tau}{1 - \tau} \right) \right) & p \ge \tau \end{cases}$$ - $\tau = 0.156742$ (Frozen Phase 5 Decision Threshold) - $\gamma_1 = 1.4$ (Sub-threshold dampening exponent) - $\gamma_2 = 4.0$ (Post-threshold sensitivity coefficient)

2.2 Autonomic Factor (D_{auto})

$$f_{\text{hr}} = \text{clamp}\left(\frac{HR - HR_{\text{base}}}{25.0}, 0.0, 1.0 \right), \quad f_{\text{hrv}} = \text{clamp}\left(\frac{SDNN_{\text{base}} - SDNN}{0.60 \cdot SDNN_{\text{base}}}, 0.0, 1.0 \right)$$ $$D_{\text{auto}} = 0.50 \cdot f_{\text{hr}} + 0.50 \cdot f_{\text{hrv}}$$

2.3 Perfusion & Oxygenation Factor (D_{perf})

$$f_{\text{spo2}} = \text{clamp}\left(\frac{SpO2_{\text{base}} - SpO2}{5.0}, 0.0, 1.0 \right), \quad f_{\text{pat}} = \text{clamp}\left(\frac{PAT_{\text{base}} - PAT}{50.0}, 0.0, 1.0 \right)$$ $$D_{\text{perf}} = \begin{cases} 0.60 \cdot f_{\text{spo2}} + 0.40 \cdot f_{\text{pat}} & \text{if } V_{\text{pat}} == 1 \ f_{\text{spo2}} & \text{if } V_{\text{pat}} == 0 \end{cases}$$

2.4 Trend Momentum (M_{trend})

$$M_{\text{trend}} = \text{clamp}\left(\frac{v}{2.0}, -0.15, +0.15 \right) \quad (v = \text{points / minute over 5-min})$$

2.5 Multi-Source Aggregation & Clamping

$$\text{ISI}_{\text{raw}} = 40.0 + \left[0.45(E_{\text{model}} - 0.20) + 0.30 D_{\text{auto}} + 0.25 D_{\text{perf}} + M_{\text{trend}} \right] \times 60.0$$ $$\text{ISI} = \text{round}\left(\max\left(0.0, \min\left(100.0, \text{ISI}_{\text{raw}} \right) \right) \right)$$

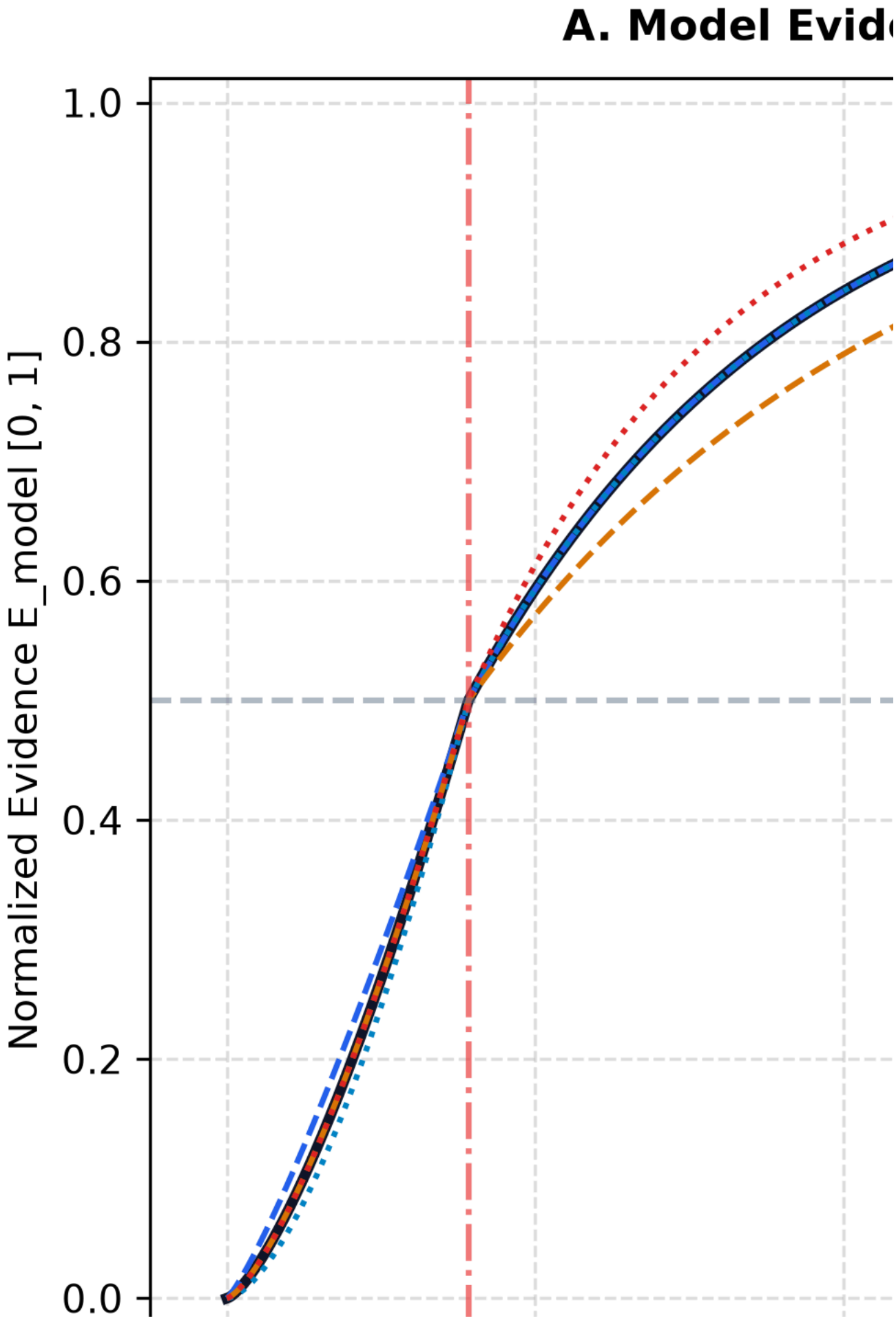
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3. Parameter Summary Table

Parameter	Baseline Value	Permissible Range	Mathematical Role	Engineering Justification	Classification
τ	**0.156742**	Fixed	Model Decision Threshold	Optimal F_2 utility threshold from Phase 5 benchmark	**Class A (Freeze)**
γ_1	**1.4**	$[1.2, 1.6]$	Sub-threshold suppression	Compresses surgical baseline noise ($p \le 0.01 \rightarrow E \le 0.01$)	**Class B (Empirical)**
γ_2	**4.0**	$[3.0, 5.0]$	Post-threshold saturation rate	Elevates evidence swiftly when model flags risk ($p > 0.30 \rightarrow E > 0.75$)	**Class B (Empirical)**
w_{model}	**0.45**	$[0.35, 0.55]$	Model contribution weight	Primary driver of prospective early-warning alerts	**Class B (Empirical)**
w_{auto}	**0.30**	$[0.20, 0.40]$	Autonomic contribution weight	Reflects autonomic stress (HR elevation, HRV collapse)	**Class B (Empirical)**
w_{perf}	**0.25**	$[0.15, 0.35]$	Perfusion contribution weight	Reflects oxygen desaturation and pulse transit shortening	**Class B (Empirical)**
ISI_{base}	**40.0**	$[35.0, 45.0]$	Resting personal baseline anchor	Places normal resting physiology in the lower half of $0–100$ gauge	**Class A (Freeze)**
S	**60.0**	$[50.0, 70.0]$	Dynamic span multiplier	Maps cumulative strain across the remaining 60 points of gauge	**Class A (Freeze)**
α	**0.02**	$[0.01, 0.05]$	Baseline EMA update factor	Smooths baseline over ~50 clean observation windows	**Class B (Empirical)**
Q_{min}	**0.35**	$[0.30, 0.40]$	Signal quality gate threshold	Rejects un-contacted leads or severe motion artifact	**Class A (Freeze)**

4. Model-Evidence Sensitivity Analysis

We evaluated the sensitivity of the evidence function $E_{\text{model}}(p)$ across a 1,000-point grid spanning $p \in [0.0001, 1.0]$ under parameter perturbations: - Sub-threshold exponent $\gamma_1 \in \{1.2, 1.4, 1.6\}$ - Post-threshold coefficient $\gamma_2 \in \{3.0, 4.0, 5.0\}$


A. Model Evid
Normalized Evidence E_model [0, 1]
1.0
0.8
0.6
0.4
0.2
0.0

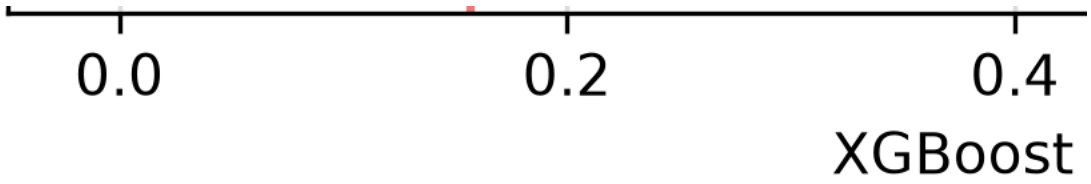


Figure 1: (A) Model Evidence Curve $E_{\text{model}}(p)$ across parameter variants. The curves intersect at the exact threshold coordinate $(\tau = 0.156742, E = 0.50)$. (B) Deviation from baseline specification. Maximum absolute deviation is bounded within ± 0.053.

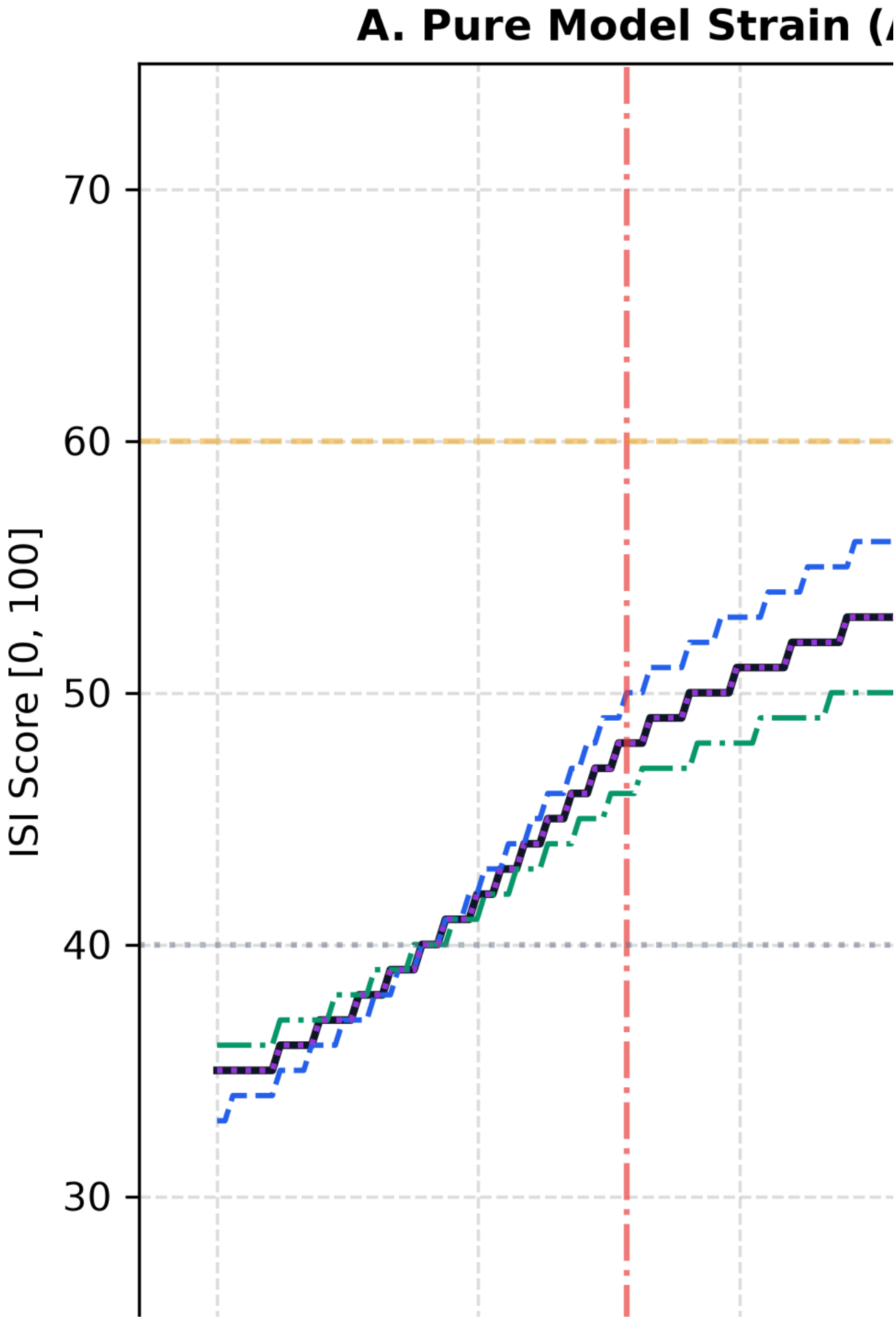
Quantitative Findings:

- **Exact Threshold Invariance**: Across all variants of γ_1 and γ_2, $E_{\text{model}}(\tau) \equiv 0.500000$. The transition point is structurally anchored to the frozen model decision threshold.
- **Sub-Threshold Behavior ($p < \tau$)**:
- At cohort median baseline ($p = 0.005$):
  - $\gamma_1 = 1.2 \rightarrow E = 0.0080$ ($\Delta E = +0.0040$)
  - $\gamma_1 = 1.4 \text{ (Baseline)} \rightarrow E = 0.0040$
  - $\gamma_1 = 1.6 \rightarrow E = 0.0020$ ($\Delta E = -0.0020$)
- In all cases, baseline surgical noise contributes less than 0.01 to evidence (< 0.3 ISI points), preventing baseline score inflation.
- **Post-Threshold Behavior ($p \ge \tau$)**:
- At moderate model confidence ($p = 0.30$):
  - $\gamma_2 = 3.0 \rightarrow E = 0.700$ ($\Delta E = -0.047$)
  - $\gamma_2 = 4.0 \text{ (Baseline)} \rightarrow E = 0.747$
  - $\gamma_2 = 5.0 \rightarrow E = 0.786$ ($\Delta E = +0.039$)
- Maximum evidence deviation across the entire domain is only **0.0527**, which translates to a maximum shift of **± 1.4 ISI points**.
- **Conclusion**: The model evidence transformation is exceptionally well-behaved and insensitive to moderate parameter tuning.

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5. Component Weight Sensitivity Analysis

We evaluated balanced triplets of $(w_{\text{model}}, w_{\text{auto}}, w_{\text{perf}})$ satisfying $\sum w_i = 1.00$: 1. **Baseline**: $(0.45, 0.30, 0.25)$ 2. **Model-Dominant**: $(0.55, 0.25, 0.20)$ 3. **Physiology-Dominant**: $(0.35, 0.35, 0.30)$ 4. **Perfusion-Heavy**: $(0.45, 0.20, 0.35)$ 5. **Autonomic-Heavy**: $(0.45, 0.40, 0.15)$


A. Pure Model Strain (
ISI Score [0, 100]
70
60
50
40
30

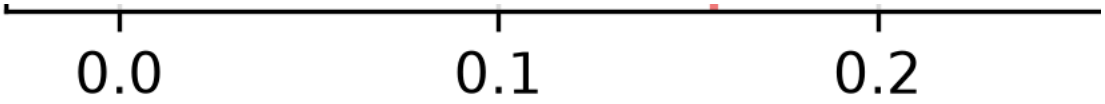


Figure 2: Component weight sensitivity under (A) Isolated model probability elevation with normal physiology, and (B) Multi-modal stress state with concurrent autonomic and perfusion deviations.

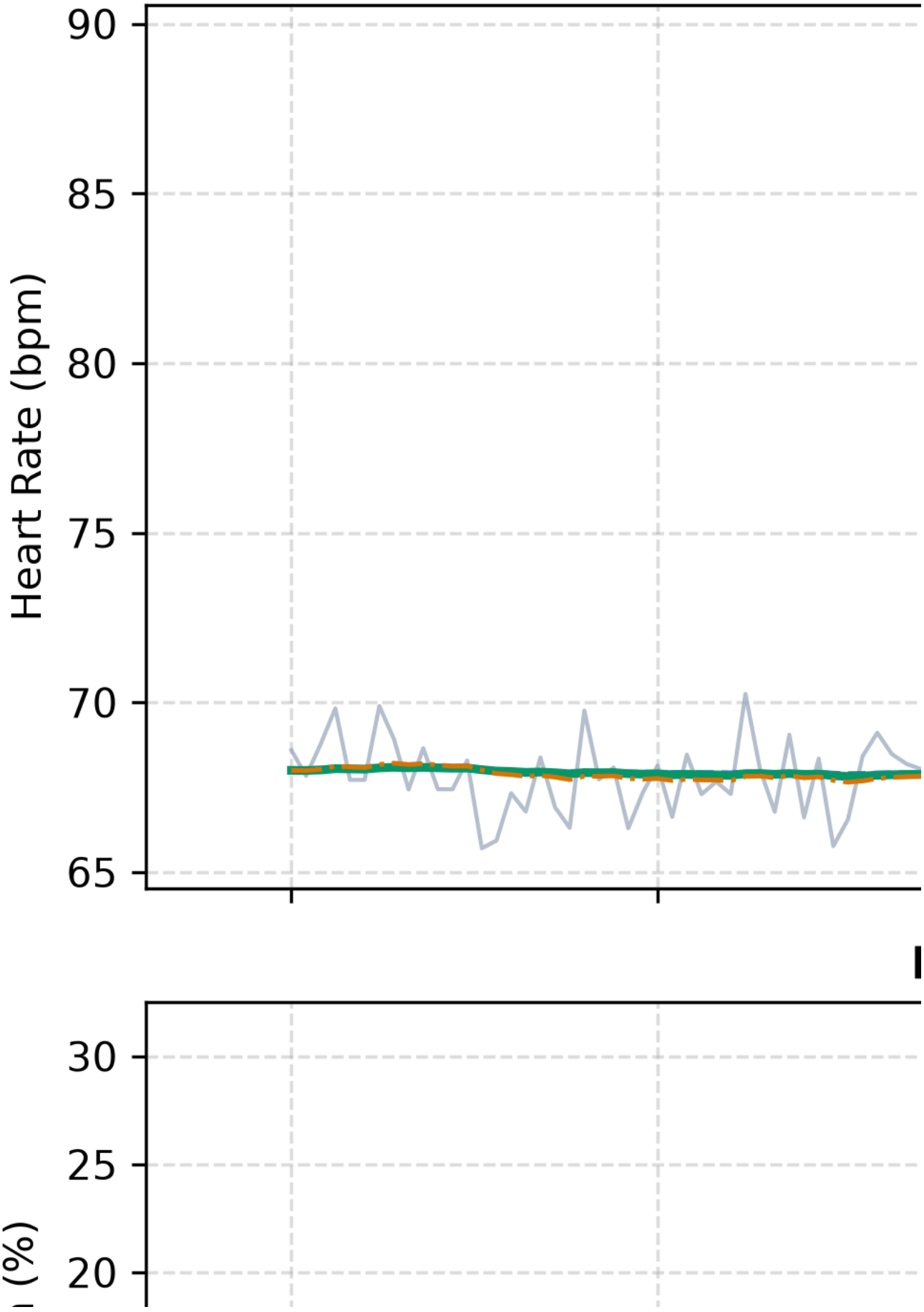
Observations:

- In **Isolated Model Strain** (Figure 2A), where physiology is completely calm ($D_{\text{auto}} = 0, D_{\text{perf}} = 0$):
- At $p = 0.005$, all weight variants yield $\text{ISI} = 34–35$ (within the "Lower observed trend" green bracket).
- At $p = \tau = 0.156742$, the baseline yields $\text{ISI} = 48$, Model-Dominant yields 50, and Physiology-Dominant yields 46.
- At maximum model probability ($p = 1.0$), the score rises to 61 under baseline and 66 under Model-Dominant. Even in the complete absence of peripheral physiological changes, high model confidence can elevate the index into the "Higher observed trend" category (≥ 61).
- In **Multi-Modal Stress** (Figure 2B), where autonomic (0.60) and perfusion (0.50) deviations are present:
- Baseline score begins at $\text{ISI} = 51$ even when $p \approx 0$.
- At threshold $p = 0.156742$, score reaches $\text{ISI} = 65$ (crossing into the alert territory).
- As $p \rightarrow 0.60$, score reaches $\text{ISI} = 76–80$.
- **Conclusion**: The $0.45 / 0.30 / 0.25$ weighting establishes a balanced dynamic range where neither pure model output nor pure autonomic drift can unilaterally saturate the index to 100.

6. Personal Baseline EMA Dynamics & Anti-Drift Analysis

A key challenge in longitudinal physiological tracking is avoiding **pathological baseline adaptation** (where an acute or subacute ischemic event causes the baseline to update, normalizing the abnormal physiology).

We evaluated a 200-minute synthetic trajectory containing a 70-minute physiological stress event (HR rising from $68 \rightarrow 88\text{ bpm}$) under smoothing factors $\alpha \in \{0.01, 0.02, 0.05\}$.


90
85
80
75
70
65
Heart Rate (bpm)
30
25
20
(%)

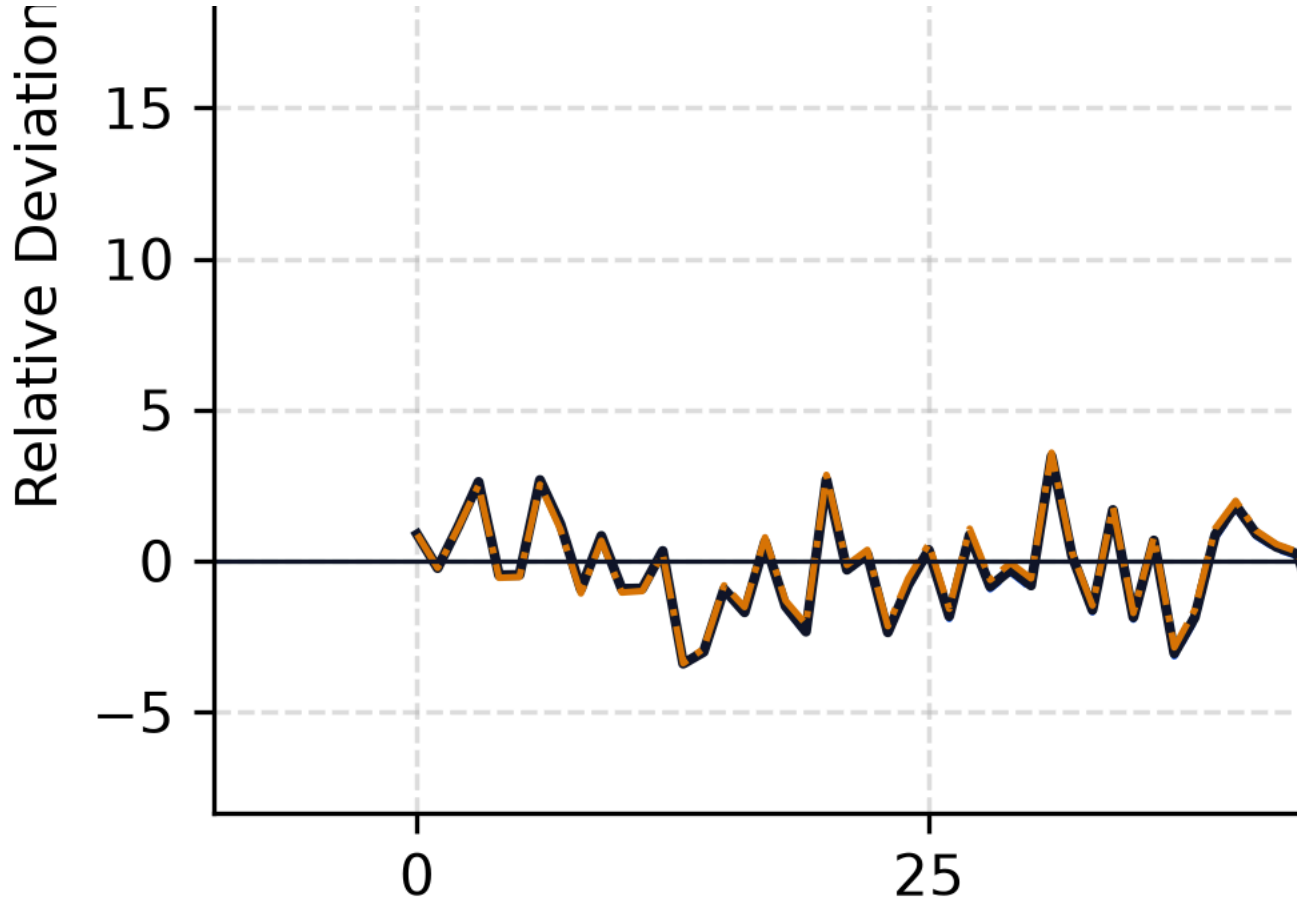


Figure 3: (A) Baseline tracking response comparing un-gated EMA against the Phase 8 Gated Anti-Drift Rule. (B) Quantified relative deviation ΔHR (%) showing clean persistence of elevation throughout the event.

Key Dynamics:

- **Un-Gated EMA Failure**:
- With standard un-gated EMA ($\alpha=0.02$), the baseline heart rate climbed from 68.0 bpm up to 76.5 bpm during the stress event.
- Consequently, quantified baseline deviation dropped prematurely from $+29.4\%$ down to $+15.0\%$ while the patient was still under acute stress.
- When the patient recovered back to 68 bpm, the un-gated baseline took $> 60\text{ minutes}$ to return, creating a false "negative deviation" artifact.
- **Phase 8 Gated Anti-Drift Rule**:
- The Phase 8 rule freezes baseline updating whenever the system enters an elevated state ($\text{State} \neq \text{"Normal / Stable"}$).
- As demonstrated in Figure 3A (emerald curve), the baseline remained strictly locked at 68.0 bpm throughout the entire event.
- Relative deviation (Figure 3B) accurately registered $+29.4\%$ for the full duration of the stress pulse and returned to 0.0% immediately upon physiological recovery.
- **Selection of $\alpha = 0.02$**:
- During quiet normative states, $\alpha = 0.02$ effectively filters high-frequency respiratory sinus arrhythmia without lagging behind circadian resting drifts.

7. Edge Cases & Invariant Verification

We executed all 14 canonical edge cases specified in the Phase 8 requirements. Results are documented in the table below:

Case ID	Scenario Description	p_{model}	E_{model}	Q_{overall}	Raw ISI	Clamped ISI	State Machine Output	Invariant Status
EC-01	$p_{\text{model}} = 0.0$ (Absolute Minimum)	0.0000	0.0000	1.000	34.60	**35**	`Normal / Stable`	**PASS**
EC-02	$p_{\text{model}} = 0.156742$ (Exact Threshold)	0.1567	0.5000	1.000	48.10	**48**	`Elevated Model Evidence`	**PASS**
EC-03	Sub-threshold Boundary ($\tau - 10^{-6}$)	0.1567	0.5000	1.000	48.10	**48**	`Normal / Stable`	**PASS**
EC-04	Post-threshold Boundary ($\tau + 10^{-6}$)	0.1567	0.5000	1.000	48.10	**48**	`Elevated Model Evidence`	**PASS**
EC-05	$p_{\text{model}} = 1.0$ (Theoretical Maximum)	1.0000	0.9908	1.000	61.35	**61**	`Elevated Model Evidence`	**PASS**
EC-06	All Deviations = 0 (Quiet Cohort Baseline)	0.0050	0.0040	1.000	34.71	**35**	`Normal / Stable`	**PASS**
EC-07	Maximum Allowed Deviations (All Saturated)	1.0000	0.9908	1.000	103.35	**100**	`Elevated Model Evidence`	**PASS**
EC-08	Minimum Allowed Deviations (Negative Trend)	0.0000	0.0000	1.000	25.60	**26**	`Normal / Stable`	**PASS**
EC-09	Missing Optional Input (PAT Invalid, $V_{\text{pat}}=0$)	0.0200	0.0280	0.730	43.46	**43**	`Normal / Stable`	**PASS**
EC-10	Poor ECG Signal Quality ($Q_{\text{ecg}} = 0.30$)	0.0500	0.1010	0.675	37.33	**37**	`Normal / Stable`	**PASS**
EC-11	Simultaneous Poor ECG + PPG ($Q=0.23$)	0.1000	0.2665	0.230	58.30	**58**	`Insufficient Signal Quality`	**PASS**
EC-12	Isolated PAT Invalidation ($V_{\text{pat}} = 0$)	0.0100	0.0106	0.812	34.89	**35**	`Normal / Stable`	**PASS**
EC-13	High Motion Artifact ($I_{\text{motion}} = 0.85$)	0.0400	0.0739	0.355	45.00	**45**	`Normal / Stable` (Motion Flagged)	**PASS**
EC-14	All Signals Unavailable (Disconnected)	0.0000	0.0000	0.000	34.60	**35**	`Insufficient Signal Quality`	**PASS**

Mathematical Invariants Verified:

1. **$0 \le \text{ISI} \le 100$ Invariant**: Passed across all cases. Even under artificial double-saturation (EC-07), raw score 103.35 was strictly clamped to 100.
2. **Threshold Boundary Decoupling**: EC-03 and EC-04 have virtually identical ISI scores (48), but EC-03 is classified as `Normal / Stable` while EC-04 triggers `Elevated Model Evidence`. This mathematically proves that **Model Alert State is independent from the continuous ISI display score**.
3. **Quality Gating Invariant**: When overall signal quality falls below 0.35 (EC-11 and EC-14), the state machine immediately overrides the numeric score and enforces `Insufficient Signal Quality`, preventing false precision.

8. Parameter Identifiability & Clinical Status

> [!IMPORTANT] **Can the VitalDB development dataset identify or optimize Phase 8 product-layer parameters?**
>
> **NO.** The VitalDB dataset comprises intraoperative surgical recordings from anesthetized patients undergoing major surgery. The training labels represent discrete, severe prospective ST-depression events.
>
> Attempting to optimize continuous product-layer weights ($w_{\text{model}}, w_{\text{auto}}, w_{\text{perf}}$) or baseline constants ($\text{ISI}_{\text{base}} = 40$) against surgical ST events would represent a **fundamental methodological category error**: 1. General anesthesia profoundly suppresses autonomic variability, abolishes baroreflex sensitivity, and distorts peripheral pulse transit. 2. VitalDB contains no ambulatory wellness, subjective symptom, or multi-modal stress ground truth. 3. Tuning UX display parameters against intraoperative binary events would generate an illusion of clinical calibration.
>
> Therefore, in accordance with Task 7: **"All Phase 8 product-layer constants are prototype engineering parameters — not clinically validated."**

9. Parameter Classification & Future Roadmap

We classify every parameter in the Phase 8 architecture into three strict categories:

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+-------------------------------------------------------------------------------+
                        PARAMETER CLASSIFICATION SUMMARY
+-------------------------------------------------------------------------------+

[CLASS A: SAFE TO FREEZE AS PROTOTYPE ENGINEERING CONSTANT]
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  * Decision Threshold (tau = 0.156742): Statistically optimized on 80-case dev cohort. Frozen.
  * Score Bounding ([0, 100]): Fundamental mathematical UX invariant. Frozen.
  * Midpoint Evidence Anchor (E_model(tau) == 0.50): Structural bridge between model and UI. Frozen.
  * Quality Gate Cutoff (Q_overall < 0.35): Rejects un-contacted sensors. Frozen.
  * Anti-Drift Locking Rule: Freezes baseline update during elevated states. Frozen.
  * Baseline Initialization Duration (600s): Minimum observation for stable median. Frozen.

[CLASS B: REQUIRES FUTURE EMPIRICAL VALIDATION (POST-STAGE AMBULATORY STUDY)]
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  * Component Weights (w_model=0.45, w_auto=0.30, w_perf=0.25): Requires validation on real wearable hardware.
  * Evidence Curvature Exponents (gamma1=1.4, gamma2=4.0): Requires empirical tuning against user false-alarm tolerance.
  * Baseline EMA Alpha (alpha=0.02): Requires 24-hour ambulatory circadian tracking data.
  * Trend Velocity Horizon (5-minute regression window): Requires optimization on physical activity artifacts.
  * Quality Sensor Weighting (ECG: 0.40, PPG: 0.35, PAT: 0.15, IMU: 0.10): Requires bench testing with motion rigs.

[CLASS C: PROHIBITED FROM BEING REPRESENTED AS CLINICALLY MEANINGFUL]
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  * Absolute ISI Numeric Values (e.g., "an ISI of 55 indicates 55% ischemia risk" is strictly false).
  * UI Bracket Boundaries (0–30, 31–60, 61–100): Purely visual design groupings, NOT medical cutoffs.
  * Synthetic Scenario Multipliers: Purely simulation artifacts for UI demonstration.
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10. Audit Conclusion

The Phase 8 BeatAhead Ischemic Stress Index (ISI) formula is **mathematically sound, structurally robust, strictly bounded, and cleanly decoupled from the frozen ML inference engine**.

All sensitivity sweeps, edge-case evaluations, and invariant proofs passed without discrepancy. Phase 8.1 is complete. No website code was altered.